1 **(a)** centripetal force is provided by gravitational force
$mv^2 / r = GMm / r^2$
hence $v = \sqrt{(GM / r)}$ A0

(b) **(i)** E_K ($= ½mv^2$) $= GMm / 2r$ B1 **[1]**

(ii) $E_P = - GMm / r$ B1 **[1]**

(iii) $E_T = - GMm / r + GMm / 2r$ C1
$= - GMm / 2r$. A1 **[2]**

(c) **(i)** if E_T decreases then $- GMm / 2r$ becomes more negative
or $GMm / 2r$ becomes larger M1
so r decreases A1 **[2]**

(ii) $E_K = GMm / 2r$ and r decreases M1
so (E_K and) v increases A1 **[2]**

2 **(a)** e.g. fixed mass/ amount of gas